



Micro-mechanical analysis on random RVE size and shape in multiscale finite element modelling of unidirectional FRP composites

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ARTICLE INFO

Keywords:

Multiscale Modelling
Representative Volume Elements
Unidirectional Composites
Microstructure
Uncertainty Quantification

ABSTRACT

Developing a high-fidelity stochastic multi-scale modelling framework that can capture the structure–property relationship in a computationally efficient manner is of interest to the structural engineers working with composite structures. This work revisits the problem of defining representative volume elements (RVEs) for performing a combined computational homogenisation-dehomogenisation procedure to estimate critical material allowables of unidirectional fibre reinforced polymer (FRP) composites. Particularly, the size and shape effects of 3D RVEs are investigated by performing Monte Carlo simulation of RVE arrangements to analyse the statistical convergence of their geometric features and the predicted structural properties in comparison with commonly used unit-cell models. For the microstructural features adopted in this work, it has been demonstrated that appropriately sized RVE models in two different aspect ratios could be considered as 3D statistical volume elements (SVEs) to efficiently capture the properties of interest. Attention has also been drawn to the implications of domain shapes and aspect ratios when small SVE models are used in the modelling framework for regulating computational power.

1. Introduction

Structural property prediction in composite materials is an area of research that receives constant attention. This is due to the complexities in understanding their process-structure–property-performance relationship and devising appropriate analytical/numerical tools and methods to predict them in a practicable manner. Particularly, the widespread efforts [1] in the recent years towards benchmarking the damage modelling practices of unidirectional (UD) fibre reinforced polymer (FRP) composite laminates have indicated the importance of considering multi-scale structural features and establishing a relationship between the constituent microstructure and the composite structure. As such, so-called multiscale modelling methods of FRP composites often include three length scales viz. the macroscopic, mesoscopic, and microscopic levels where different geometric features are present [2–4]. Such multiscale modelling methods differ in the consideration of sub-length scale features in their property prediction for both micro-to-macro homogenisation and macro-to-micro dehomogenisation/localisation approaches [5]. A challenging aspect of these multi-scale finite element modelling approaches involves balancing the computational

load and accuracy that comes with high-fidelity models. When coupled with upcoming real-time production and in-service monitoring frameworks of composite structures like Digital Twins [6] that require the simultaneous update of these high-fidelity multi-scale models, there is an ever-growing need to optimise them.

The fundamental tenet to the framework of both the multiscale modelling approaches is the concept of representative volume elements (RVEs) which was originally put forward by Hill [7] to analytically determine homogenised effective properties. This is also predominantly applied in the computational homogenisation based multiscale modelling approaches to design the composite structure for stiffness. The effective response of the finite element model of an RVE is computed as field averaged value which is then considered to be an effective value of the equivalent homogenised UD lamina. In the dehomogenisation/localisation approaches of designing for strength, Hill's definition of RVE is extended to include the distribution of stresses or strains within the phases of the RVE. Among the example works [8–12] under this category, some consider extreme values within the phases [8–10] whereas others consider phase-average statistics [11,12] of stresses/strains to determine the initiation of damage within a UD lamina

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<https://doi.org/10.1016/j.compstruct.2021.115081>

Received 6 August 2021; Received in revised form 20 October 2021; Accepted 4 December 2021

Available online 10 December 2021

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according to a defined criterion. Irrespective of the design objective, researchers in the past have simplified the RVE to be a unit cell (UC) by assuming certain repeating patterns in the fibre arrangement such as hexagonal, square, etc. [13–17] as shown in Fig. 1 to mainly overcome the computational challenges. However, such unit-cell modelling approaches have been shown [18] to provide an acceptable measurement of effective (spatially averaged) elastic properties in homogenisation approaches but local damage onset and ultimate failure prediction in dehomogenisation approaches can be underestimated as they ignore the underlying variability in microstructure and material properties. It has since been well established [19–25] that both the effective elastic properties (and hence stiffness) and the local field values that determine the microstructural damage behaviour (and hence strength) are a function of the RVE which should be formulated appropriately to be accurate and internally consistent in performing a reliable multi-scale analysis.

Several microstructure descriptors have been proposed in the literature for statistically quantifying the micro-scale characteristic features influencing the global structural behaviour of UD FRP composites so that they can be used to appropriately generate/reconstruct RVEs with statistical equivalence to real microstructures. In the broader field of computational mechanics, such commonly used descriptors can be classified [26] into physical geometry descriptors and statistical functions. Geometric descriptors are physically meaningful parameters with deterministic values or statistical distributions that describe the constituent phases - e.g., volume fraction, number of fibres, fibre shape, fibre dimensions, inter-fibre distance, micro-voids/defects, Voronoi polygon areas [27], nearest neighbour distances and nearest neighbour orientations. Statistical functions are spatial correlation functions that describe the spatial relationship between the constituent phases - e.g., Ripley's K-function (second-order intensity function) [28,29], pair distribution function [30], radial distribution function [31,32] and two-point correlation function [33,34] (generally known in the literature as n-point correlation function or n-point probability function or n-point spatial correlation function). It should be highlighted that only the physical descriptors are widely being used to algorithmically generate RVEs in the composites community. Among the features mentioned, the commonly used metrics for measuring the size of RVEs (as a proxy for computational load) representing UD composites include the number of fibres ($N_f = V_f \delta^2 / \pi$) enclosed in the RVE domain of certain fibre volume fraction (V_f) or alternatively, the ratio ($\delta = L/r$) of RVE characteristic length (L) to the radius (r) of the fibres.

Many have looked at studying the size of RVE models with random fibre arrangements in UD laminates over the recent years in the context of homogenised effective properties (full set of orthotropic properties or

only selected in-plane parameters) [15,18–20,23,24,35–39] and/or localised damage initiation [21,22,25,40–50]. These works can be grouped based on the consideration of: (i) source of RVEs viz. algorithmically generating RVEs [27,41,51], reconstructing RVEs from real micrographs [22,23,52]; (ii) microstructural features viz. fibre shape and dimensions [19,37], micro-voids [42,43,53]; (iii) constituent materials viz. isotropic fibre materials like glass [22,38,40–42,47,49,51,53] or transversely isotropic fibres like carbon [21,23,52], perfect bonding between fibre and matrix [15,21,23,35,38,40,41,47,49,51–53], explicitly modelling fibre–matrix interface and debonding [21,22,37,42,43,48,50]; (iv) finite element model setup viz. application of periodic boundary conditions (PBC) or otherwise, plain strain assumption to model the RVE using 2D elements [21,39–42,47,49,51–53] or a full 3D RVE using continuum elements [20,22,23,38]; (v) damage/failure criteria viz. micro-mechanics of failure (MMF) [15], Strain-Invariant Failure Theory (SIFT) [50], Multi-continuum Theory (MCT) [12] and critical strain energy density [41,49,51,53] among others. Most of the above works have arrived at different conclusions in terms of the size of the RVE domain¹ as expected due to their inherent differences highlighted earlier. The proposed δ values of RVEs range from 9 [23] to as high as 60 [49]. In terms of N_f , it ranges from 30 fibres [42] up to 200 fibres [41] inside the RVE domain of generally 60% fibre volume fraction.

While certain simplifications to the RVE models are appropriate under specific conditions such as using generalised 2D plain-strain element models for measuring only the in-plane effective properties and transverse matrix damage modes, a generalised high-fidelity modelling approach that can be applied to a whole host of scenarios involving 3D stress/strain states should ideally consider a full 3D RVE with all relevant microstructural features, and physically-based constituent material-level failure criteria. Consequently, there is clearly an issue of convergence before a specific RVE size is adopted, which may require significant effort. Particularly when concerned with the damage initiation of orthotropic materials like composites, this can become very significant as the large RVE models required to establish statistical convergence could be intractable. Alternatively, it has been shown in the literature [23,26,51,54] that one can also build and analyse an ensemble of reasonably smaller RVE models, called Statistical Volume Elements (SVEs), to get the converged statistical range of the parameters of interest. Based on the design philosophy employed, the composite structure can then be designed to ensure stiffness or strength requirements are met without exceeding the critical value as observed from experiments. This provides an opportunity for finding an appropriate balance between the computational cost associated with modelling large RVE models and the simplicity available from unit-cell models from a practical engineering design perspective by arriving at a pseudo-random SVE arrangement such as the one illustrated in Fig. 1. Several different algorithms [15,23,27,35,38,39,41,53,55–59] have been proposed over the years on generating such random RVE models with varying degrees of statistical equivalence to real composite specimens. Some differ in their ability to directly control the periodicity of the microstructure at boundaries which aid in the application of periodic boundary conditions for numerical analysis. Almost all approaches in the literature also consider the shape of the RVE to be square and fibres to be circular and generate RVE models of varying dimensions to achieve the desired volume fraction with constant fibre diameter. While some have investigated the shape effects of fibres [19,37], to the best of the authors' knowledge, there are no studies in the literature that have examined if the aspect ratio of the random RVEs influence the predictions as they do in the case of unit cells.

In the light of the above literary works and background, the primary focus of this paper is to investigate random 3D RVEs in comparison with

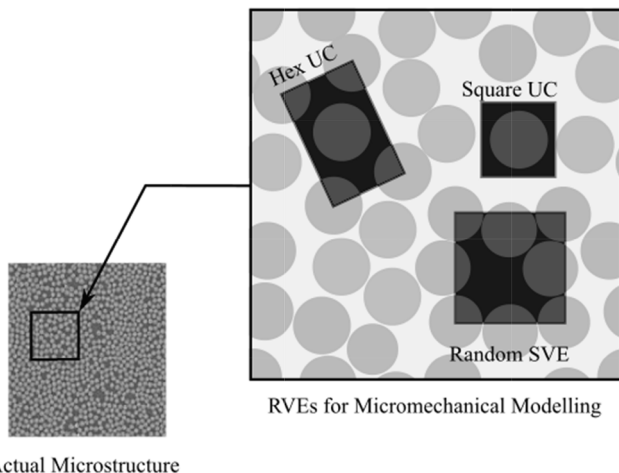


Fig. 1. Graphical illustration of various micromechanical RVE models for a unidirectional lamina.

¹ in some cases, even concluding unit cells as RVEs [15]

unit cells to propose a set of pseudo-random 3D RVEs (referred to as SVEs) sized in terms of the minimum number of fibres and aspect ratios. The requirements for such pseudo-random SVEs to provide an acceptable prediction of effective elastic properties and damage onset while ensuring convergence of microstructure descriptors are also examined. With regards to studying the damage or failure within composite materials, a brief discussion on the expressions of damage/failure criteria used to determine the occurrence, or lack thereof, damage within the constituent materials is often required. Various expressions have been proposed for constituent failure criteria based on the direct experimental observations on virgin constituents or derived from laminated plies with constrained constituents, some of which have been mentioned earlier. Per se, Onset Theory, formerly known as Strain Invariant Failure Theory (SIFT), [60] is used as an underlying dehomogenisation based criterion to examine the effect of RVE size and shape on the predictions of damage onset within the matrix [6,11,15]. The findings of this work could be of interest to the practitioners of this approach as the existing works [8,9,12,15] have mainly proposed unit cell models. The methodology of arriving at a set of SVEs recommended by Trias [51] is followed here and hence may also add value to the wider body of literature investigating this problem.

The paper is structured as follows. Section 2 provides a brief overview of the steps involved in generating finite element models of random 3D RVEs and the extensions to incorporate random 3D RVE models in dehomogenisation based damage prediction methodology of Onset Theory. Section 3 presents the statistical characterisation of random fibre arrangements to understand the convergence of fibre arrangements and Section 4 illustrates the effect of fibre arrangements on the homogenised elastic properties, dehomogenised strain invariants and local damage initiation predictions to establish underlying correlations. The brief discussion in Section 5 then proposes a set of SVE models that can be used in building a computationally efficient yet reliable multiscale finite element modelling framework for UD FRP composites.

2. Modelling methods

2.1. Generation of Pseudo-random RVEs

To investigate the shape effects of SVEs, the geometrical shapes of the random RVE domains have been inherited from unit cells namely the square and hexagonal to consider the packing constraints caused by different aspect ratios of the RVEs. To limit the scope of the study, fibres are assumed to be circular in shape with constant fibre diameter inside any single RVE ignoring the change in fibre centroid locus along the fibre direction. The resin domain is modeled without voids or pre-cracks and a perfect bonding interface is considered between fibres and matrix. While the change in fibre centroid locus along the fibre direction is inconsequential for the matrix behaviour, explicitly detailing the other microstructural features may further influence the critical size of the RVE and will be a subject of future study. Also, for convenience in comparing the microstructural features, this work conversely defines the RVE domain as a fixed window of desired aspect ratio that can potentially be observed with matching boundaries while the fibre diameter is scaled to achieve the desired volume fraction for the RVE domain. The algorithm for generating random RVEs of various sizes utilised in this work is equivalent to the coin dropping approach [35] in which randomly spaced, non-overlapping circles in a plane within a domain are sampled and then extruded to the desired dimension in the normal direction to the plane to obtain parallel cylinders of fibres.

The RVEs considered in this work are generated by setting four target parameters namely the fibre volume fraction (V_f), number of fibres within the RVE (N_f), aspect ratio of the RVE and the minimum inter-fibre clearance. The aspect ratio is set to be 1 and $\sqrt{3}$ for the square and hexagonal RVEs resulting in cross-section areas of 1 sq. unit and 1.732 sq. unit, respectively. The minimum inter-fibre distance is

restricted to 2% of the fibre diameter to avoid touching fibres causing poor mesh quality between the fibres. 320 random RVE models (2 RVE aspect ratios, 16 RVE sizes with number of fibres ranging from 2 to 28 fibres, 10 fibre arrangement trials for each) are generated and analysed for this study, along with the square and hexagonal configuration of unit cells. A sample ensemble of the random RVEs generated for different N_f with 60% V_f is shown in Fig. 2.

The fibre centre coordinates and the RVE dimensions are exported from the Python language implementation of the algorithm and used in Abaqus CAE to build the 3D finite element models of the RVEs by extruding in out-of-plane direction. The models are meshed using six-node linear wedge elements (C3D6) with the appropriate seed size to ensure mesh convergence and matching mesh faces. Periodic boundary conditions are enforced by applying Equation constraints in Abaqus using its Python scripting interface similar to the approach in [61]. Six canonical mechanical load cases and a thermal condition are applied to the RVEs from which micro-mechanical enhancement matrices (MMEs) (detailed in Section 2.2) and a full orthotropic set of effective elastic properties can be obtained using the well-known concept of computational homogenisation [5,19,20,38,61]. The exaggerated deformed view of a sample square RVE is shown in Fig. 3 to demonstrate that the displacements of any set of opposite nodes on the boundary of the RVE are always equal indicating that the periodic boundary conditions are effectively realized for the RVEs. The same set of procedures is also followed for building and analysing the square and hex configurations of unit cells.

2.2. Damage modelling

Application of Onset Theory to assess the influence of random RVEs on damage onset prediction requires an understanding of the modular and hierarchical dehomogenisation procedure that allows for the evaluation of stress/strain state variables within the constituents of the composite materials. This section presents an overview of the localisation/dehomogenisation technique used in Onset Theory with a specific focus on the modifications required to accommodate random RVEs. The reader is referred to [9,62,63] for a detailed account on the physical basis behind the theory and its application to fibre reinforced composite materials. The over-arching framework for the prediction of damage onset in a composite structure using Onset Theory is presented in Fig. 4 as a schematic diagram. Physical testing programme involves characterising the constituent materials such as fibre and matrix and the cured laminate coupons to obtain the corresponding elastic properties. The estimated constituent properties are then used in the finite element modelling of random RVEs and unit cells to extract micro-mechanical enhancement (MME) factors by applying periodic boundary conditions. Currently, there is a lack of reliable experimental techniques to directly measure critical strain invariants from virgin resin samples subjected to similar stress/strain states that of a constrained matrix in UD lamina. Instead, coupled experimental - numerical analysis of off-axis tensile testing is recommended in which the lamina failure strain measured is used to deduce critical strain invariants. Finally, the finite element simulation of a composite structure can be performed in which the MMEs (stored as material property database files) will be used to dehomogenise the continuum strain tensors of the lamina level to the constituent level which will be used to evaluate damage onset within the constituents.

The key advantage of the dehomogenisation technique used in Onset Theory is that it decouples the numerical analysis of the microscale models from the analysis of the continuum structural model greatly reducing the computational effort. This is achieved by constructing the MME matrices that relate the strain state of a homogeneous material as modelled in macro-scale to the strain state at specific interrogation points in the RVE models as in Eq. (1). To achieve a high-resolution map of the mechanical behaviour at micro-scale, the MME matrices ($\mathbf{M}$ from

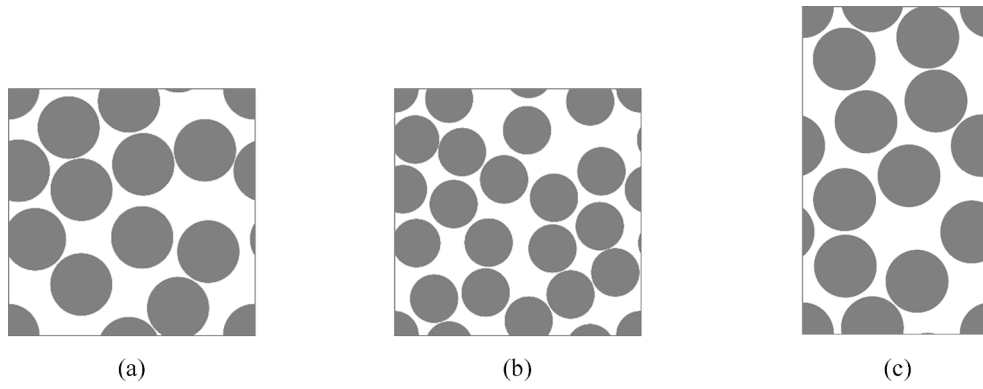


Fig. 2. Square RVE with the number of fibres equals: (a) 12, (b) 20, (c) Hexagonal RVE with 12 fibres.

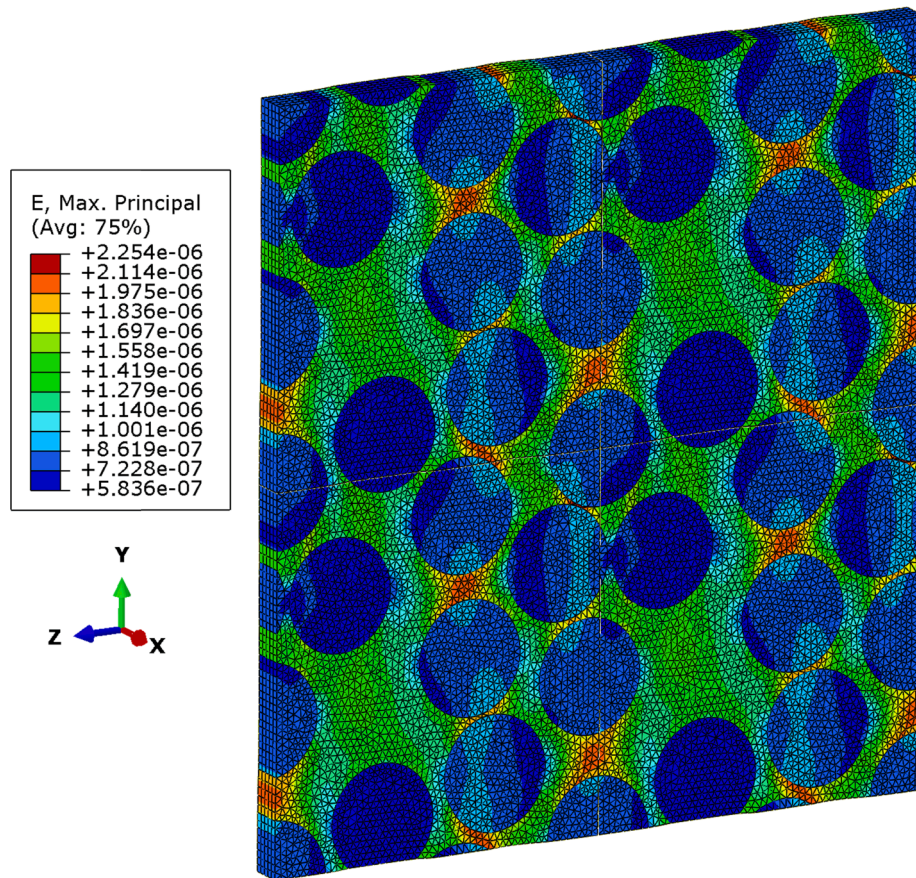


Fig. 3. A 2×2 pattern illustration of a sample square RVE under unit strain tensile loading in the Y direction with periodic boundary conditions (maximum principal strain plotted at 95 k times scaling).

six canonical mechanical load cases and $\mathbf{A}$ from thermal condition in Eq. (1) are sampled at multiple locations in the finite element model, whose index is denoted as m . These sampling (interrogation) points are chosen based on their ability to represent the local maxima within the constituent phases for any general loading condition and is of importance to the type of RVE modelled. Previous literature on dehomogenisation approaches with unit cells [8,9,12,13] has concluded that the points at Inter-fibre (IF), Interstitial (IS), Inter-fibre Interface (IFI) and Interstitial Interface (ISI) locations are representative of the matrix constituent for strength prediction as they sustain the highest strain invariants. A reasonable justification for the selection of such interrogation locations using exemplary invariant contour plots of unit cells can be found in [64]. To have a robust consideration of all the possible random

arrangements of the fibres, the transverse orientation of the RVE model with reference to the ply level coordinate system is accounted by employing tensor rotation of continuum strain tensor (${}^{\theta}\bar{\epsilon}$) about the fibre axis (x-axis in Fig. 5).

$${}^{m,\theta}\epsilon = {}^m\mathbf{M} : {}^{\theta}\bar{\epsilon} + {}^m\mathbf{A}\Delta T \quad (1)$$

The identification of such IF and IS points for irregular fibre arrangement patterns in RVEs is achieved using a 2D Voronoi Tessellation with the fibre centre locations as seeds. The IF points lie at the location directly half-way between adjacent fibres (as determined by the Delaunay Triangulation dual graph) and the IS points lie at the vertices of the Voronoi polygons. ISI and IFI interrogation points are not used in this study but could be included in a straightforward manner based on

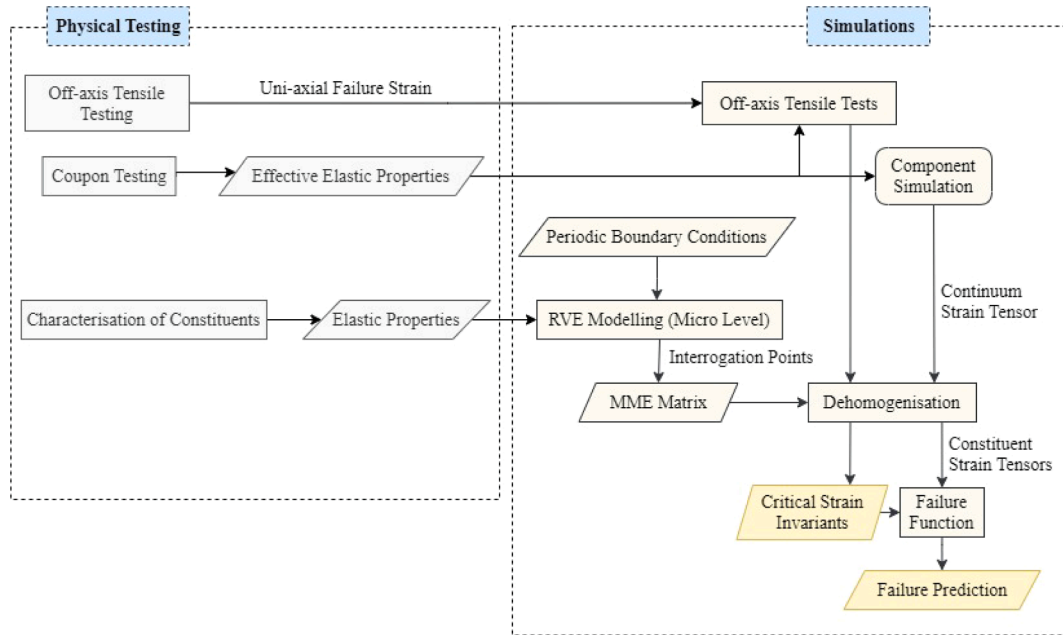


Fig. 4. Onset Theory framework to predict damage initiation of a composite structure.

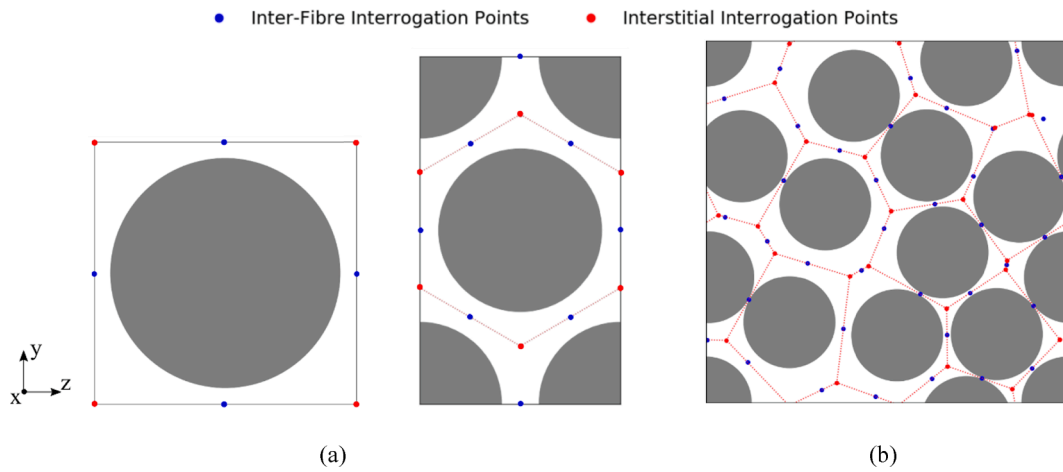


Fig. 5. Interrogation points within: (a) unit cells (square and hexagonal) (b) a sample random RVE.

similar definitions. The important feature of the 2D Voronoi Tessellation approach is that it reduces to the definition of interrogation points in unit cells as the fibre arrangement approaches regularity. The interrogation points for square and hex unit cells and an example random RVE are shown in Fig. 5.

Once the strain states at micro-level ($m, \theta \epsilon$) are obtained following the dehomogenisation step using Eq. (1), damage onset within the constituents is evaluated using a suitable criterion to get to a scalar failure index from a tensor strain field. SIFT postulates that damage initiates in matrix by two fundamental modes namely the dilatational and distortional modes which is based on the value of dilatational and distortional strain invariants estimated using Eqs. (2) and (3).

$$\epsilon_{dil}(\epsilon) \equiv J_1 = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} = \epsilon_1 + \epsilon_2 + \epsilon_3 \quad (2)$$

$$\epsilon_{dis}(\epsilon) = \sqrt{\left\{ \frac{1}{6} \left[(\epsilon_{xx} - \epsilon_{yy})^2 + (\epsilon_{yy} - \epsilon_{zz})^2 + (\epsilon_{zz} - \epsilon_{xx})^2 \right] + (\epsilon_{xy}^2 + \epsilon_{yz}^2 + \epsilon_{zx}^2) \right\}} \quad (3)$$

Each of the strain invariants corresponds to a critical material allowable. Damage (cavitation) by volumetric expansion of the matrix is predicted to initiate when the dilatational strain invariant exceeds its critical value $\epsilon_{dil,cr}$ and the damage onset by shape distortion (yielding) of the matrix initiates when the distortional strain invariant exceeds its critical value $\epsilon_{dis,cr}$, as shown in Eq. (4).

$$\max\left(\frac{\epsilon_{dil}}{\epsilon_{dil,cr}}, \frac{\epsilon_{dis}}{\epsilon_{dis,cr}}\right) \geq 1 \quad (4)$$

3. Statistical analysis of fibre arrangements

In this section, the statistical convergence of microstructural features in the generated RVEs is studied. Of the microstructure descriptors listed in Section 1, one geometric descriptor (gap ratio) and two statistical functions viz. Ripley's K function and two-point correlation function are used to evaluate the statistical convergence of microstructural features.

3.1. Gap ratio

It is widely known that the variability in the inter-fibre distance in a random RVE perturbs the local field values significantly compared to the unit cells due to the more uniform inter-fibre distances seen in the latter. To quantify this variability in random RVEs, the ratio of inter-fibre distance to fibre diameter referred to as 'gap ratio' is observed among the nearest neighbouring fibres. This normalised form of the first nearest neighbour distance is used so that the random RVEs with different sizes generated in Section 2 can be compared in a meaningful manner. Fig. 6 shows the mean and coefficient of variation (CoV) of the gap ratio in the random RVEs of square and hex aspect ratios with various RVE sizes. In both the figures, the x-axes correspond to the RVE size in terms of the number of fibres, and the colour of the lines corresponds to the aspect ratio of the models. Both the mean and CoV tend to oscillate for the smaller RVE models in both square and hex cases, but they rapidly converge for RVEs with over 11 fibres. For comparison, the mean gap ratio for square and hex unit cells are represented as red and blue dotted lines respectively, while their CoV would be zero, which is indicative of the regularity in unit cells models.

3.2. Ripley's K function

One of the salient features of random RVEs over unit cells is their ability to capture the clustering of fibres and resin-rich pockets seen in real micrographs of FRP composites. Ripley's K function, a statistical function used to assess long-range clustering/regularity or randomness in point patterns, has been widely utilised in the literature to study the fibre clustering effects of UD FRP composites. In two dimensions, Ripley's K function, as defined in Eq. (5), measures the average number of points contained in a circle of a certain search radius r , divided by the total density of points in the RVE.

$$K(r) = \frac{A}{N^2} \sum_i \sum_{j \neq i} \frac{I(d_{ij} \leq r)}{w(l_i, l_j)} \quad (5)$$

where l_i and l_j denote two generic points in the RVE and d_{ij} their distance. $I()$ is a function returning a value of 1 if the condition in brackets is true and 0 otherwise; and $w(l_i, l_j)$ is a weighing function taking the value of 1 when the circle centred in l_i and passing through l_j is entirely contained in the RVE; conversely, $w(l_i, l_j)$ returns the fraction of the circle's area contained in the RVE. As a reference for the case of a pure random distribution of points corresponding to a Poisson process, in which points are located randomly in space, the K function reduces to:

$$K_p(r) = \pi r^2 \quad (6)$$

To facilitate visualisation and easier interpretation, the residual value of the variance stabilized Ripley K function called the L function as defined in Eq. (7) is used for analysis in this paper. For a given distribution of points, if the L function is higher than zero, this indicates clustering while a value less than zero or an oscillating and non-converging behaviour indicates periodicity.

$$L(r) = \sqrt{\frac{K(r)}{\pi}} - r \quad (7)$$

To ensure consistency in the relative magnitude of the L function for increasing number of fibres, the fibre diameter and the RVE domain dimensions exported by the Python language implementation of the algorithm described in Section 2 are re-processed by setting the fibre diameter to a unit value and the RVE domain size is scaled accordingly. The plot of L function against the search radius to fibre diameter ratio (r/D) for one trial from only selected RVE sizes of 60% V_f is shown in Fig. 7 alongside the truly random Poisson process which has a value of zero. The residual function of RVEs with less than 12 fibres in both square and hex RVE shapes are oscillating indicating some degree of periodicity. The amplitude of the oscillations in residual function reduces and approaches zero for RVE models containing more than 12 fibres at large values of search radius indicating long-range randomness in such models.

3.3. 2-Point correlation function

The 2-point correlation function is a subset of the n -point statistical correlation functions that can be used to quantify the morphological relationship between different phases in the entire RVE microstructure. This is unlike Ripley's K-function which quantifies only the interaction between fibres based on the simplified representation of fibres as points. As such, the 2-point correlation function contains a lot more high-level information and has attained extensive adoption in characterising microstructures of a wide range of materials such as metallic polycrystals, co-polymers, concrete, etc. [26] but has seen little uptake in studying FRP composites.

A 1-point correlation function defines the cumulative probability of finding a constituent phase in the microstructure RVE which is equivalent to the volume fraction of that phase. The 2-point correlation function, provided in Eq. (8), defines the probability of finding phases p and p' (here fibre or matrix) at the head and tail, respectively, of a vector of length r , randomly placed in the pixel-image based representation of the microstructure (S is the total number of pixels in the image). Each pixel in the image is identified by a unique 2D position vector s .

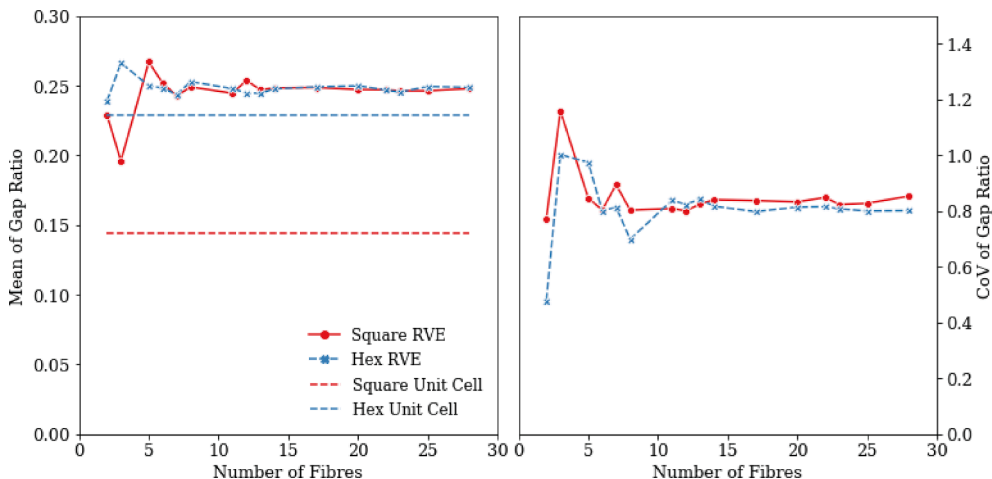


Fig. 6. Variation of mean gap ratio (left) and CoV of gap ratio (right) for RVE sizes and types.

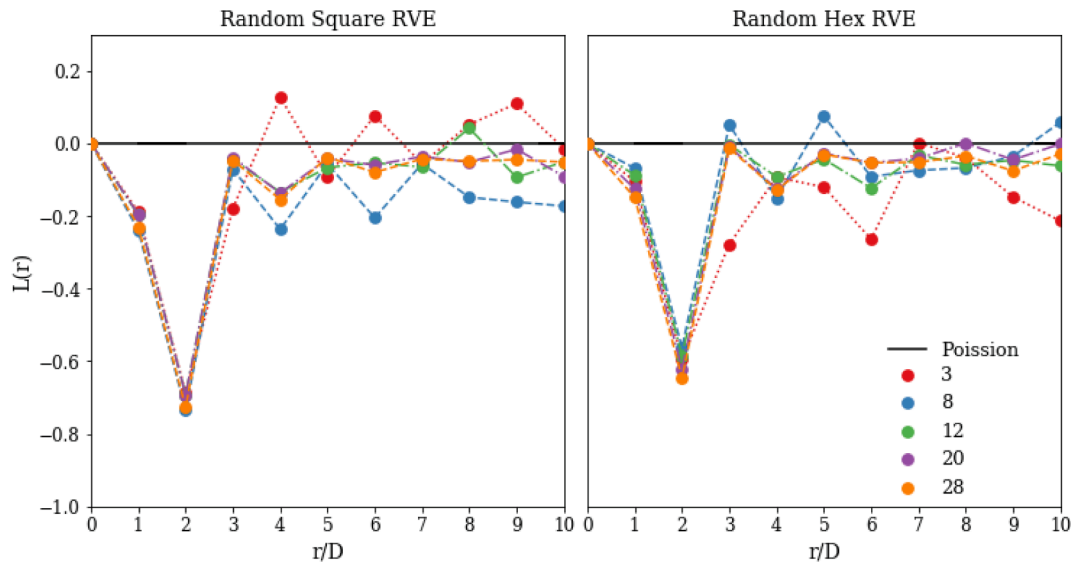


Fig. 7. Evolution of the Residual K Function for random RVEs.

$$f_r^{pp'} = \frac{1}{S} \sum_{s=1}^S m_s^p m_{s+r}^{p'} \quad (8)$$

where the microstructure function m_s^p is a function taking the value of either 1 if phase (p) is present at the spatial position (s) or 0 if otherwise. For periodic two-phase microstructures, the four possible 2-point correlation functions (two auto- and two cross-correlations) are interrelated, such that only one function is sufficient. As a result, the inter-fibre spatial auto-correlation function f_r^{FF} is used for analysing the periodic two-phase FRP composite RVEs in this work. The inter-fibre correlation f_r^{FF} for the sample RVE microstructures shown in Fig. 2 is illustrated in Fig. 8.

Niezgoda et al. [54] have proposed and demonstrated that these 2-point correlation statistics of RVEs can be used to arrive at a set of SVEs that can reasonably capture the abstract geometrical features and the inter-play of fibres that may not be captured by simply using geometry descriptors or other statistical functions. This approach has been adopted here to evaluate the statistical convergence of RVE fibre arrangements with some modifications. The original approach requires the RVEs to be obtained as subsets of a large microstructure ideally

sourced from the real specimen. However, this poses computational challenges in maintaining the periodicity of RVEs which is required to obtain matching mesh faces for enforcing periodic boundary conditions. Alternatively, by virtue of random sampling, one can reasonably assume that all the 320 algorithmically generated RVEs in Section 2 are derived from a single large micrograph of a UD specimen. Then, the average of inter-fibre correlations of all the RVEs can be defined as,

$$\bar{f}_r^{pp'} = \frac{1}{J} \sum_{j=1}^J (f_r^{pp'})_j \quad (9)$$

where, J stands for the number of RVE models (160 per aspect ratio in this study) derived from the master micrograph. A scalar measure of difference between any single RVE with that of the master micrograph can then be obtained by calculating algebraic norm of the difference in inter-fibre correlations (similar to the simple root-mean-square error) between a single RVE and that of the ensemble average. The statistical convergence of this scalar difference can then be used to identify the RVE size where a particular RVE is able to reasonably represent the higher-order statistical measures of any higher sized RVE. Fig. 9 shows

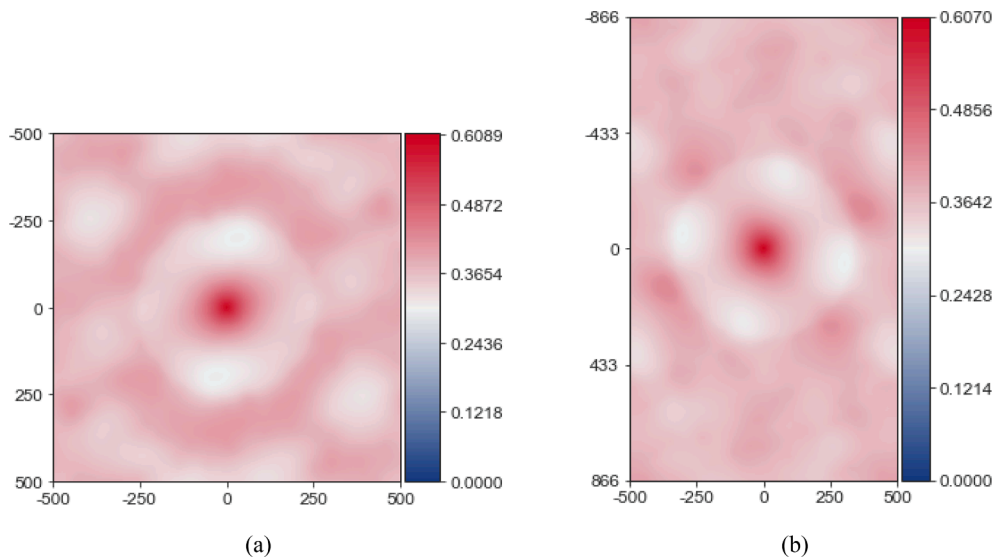


Fig. 8. 2-point auto-correlation contours of sample RVEs in Fig. 2: (a) Square ($12 N_f$), (b) Hex ($12 N_f$).

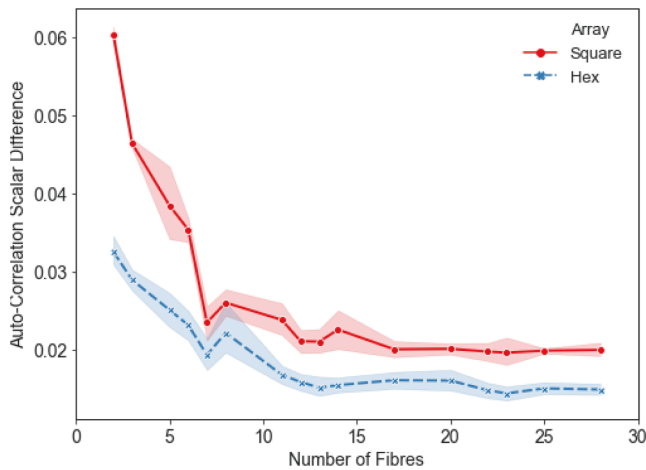


Fig. 9. Evolution of the 2-Point Auto-correlation Norm Difference for random RVEs.

the mean of the scalar difference in the random RVEs of square and hex aspect ratios for varying sizes with error band indicating the 95% confidence interval, which has reached a plateau for RVEs with over 11 fibres. It is important to note that hex RVEs consistently have marginally lower scalar error than the square RVEs signalling that any single hex RVE can sufficiently capture the underlying microstructural features than the equivalent square RVEs owing to their larger domain cross-section area.

4. Results

In this section, the convergence of the properties predicted by using different random RVE sizes and shapes is analysed by comparing the following parameters: (1) the ‘homogenised’ effective elastic properties estimated using RVEs, (2) the ‘localised’ critical strain invariants at the micro-scale following the dehomogenisation of the off-axis tensile failure strain of composite specimens, and (3) local damage initiation predictions of tensile loaded open-hole composite specimens. Also, the more fundamental comparisons in numerical estimates of these same set of parameters between RVEs and unit cells are also presented. T300 / CYCOM 970 UD prepreg is chosen as the choice of unidirectional carbon fibre/epoxy composite material system for this investigation. Elastic property values of the T300 fibre and CYCOM 970 resin used in the micromechanical modelling of random RVEs for extracting MMEs and homogenised elastic properties are given in Table A1. Failure strain tensor data of the off-axis tension tests conducted on six different ply orientations of 52 coupons as part of another study [65] are used for

Table A1
Material properties for UD T300/CYCOM970 prepreg, fibre and resin.

Property	Fibre	Resin	T300/CYCOM970 prepreg
Volume Fraction	0.6*	0.4	
E_{11} (GPa)	230 [‡]	3.44*	135*
E_{22} (GPa)	12 [‡]		7.5 [‡]
E_{33} (GPa)	12 [‡]		7.5 [‡]
G_{12} (GPa)	9.85 [‡]	1.24*	3.5*
G_{23} (GPa)	4.62 [‡]		2.76 [‡]
G_{13} (GPa)	9.85 [‡]		3.5*
ν_{12}	0.17 [‡]	0.389*	0.25 [‡]
ν_{23}	0.30 [‡]		0.45 [‡]
ν_{13}	0.17 [‡]		0.25 [‡]
α_1 (°C)	-5.40e-7 [‡]	8.06e-5*	1.00e-7 [‡]
α_2 (°C)	-2.25e-5 [‡]		2.87e-5 [‡]
α_3 (°C)	-2.25e-5 [‡]		2.87e-5 [‡]

* Directly measured quantity

‡ Measured with some uncertainty

‡ Taken from data sheet, published values or best estimates

Table A2

Critical strain tensors obtained from off-axis tension tests of T300/CYCOM970 UD lamina.

Off-axis angles (degrees)	ϵ_{11} ($\mu\epsilon$)	ϵ_{22} ($\mu\epsilon$)	ϵ_{33} ($\mu\epsilon$)	γ_{12} ($\mu\epsilon$)	γ_{23} ($\mu\epsilon$)	γ_{31} ($\mu\epsilon$)
10	8016	2509	-4060	54,994	0	0
20	3862	8860	-5440	57,057	0	0
30	1795	11,276	-5783	43,640	0	0
45	663	14,656	-6910	32,751	0	0
67	-54	13,768	-6240	12,510	0	0
90	-220	15,954	-7170	0	0	0

dehomogenisation analysis. Typical example of these tensors utilised in this paper obtained using the procedure outlined in [65] is shown in Table A2.

4.1. Effect on homogenised properties

Fig. 10 illustrates the trend in estimated mean effective properties of UD lamina using computational homogenisation of unit cells and random RVEs with varying number of fibres and aspect ratios. The average of ten trials for each RVE size is provided with error band indicating the 95% confidence interval of the average. The predictions of the analytical model by Hashin [66] is also provided for comparison. While the longitudinal Young’s modulus (E_{11}) is least affected by the RVE size or shape, the transverse modulus values (E_{22} , E_{33}) tend to converge for random RVEs with more than 11 fibres for both aspect ratios and are within Hashin’s bounds. The out-of-plane (ν_{12} , ν_{13} , G_{12} , G_{13}) and in-plane (ν_{23} , G_{23}) values of Poisson’s ratio and shear modulus are also largely varying due to fibre arrangements in the smaller RVE sizes with their values converging to the estimation of Hashin’s model for more than 11 fibres in the RVE. The transversal isotropy of the homogenised lamina in the planes perpendicular to the fibre direction is also observed for such RVEs resulting in $E_{22} \approx E_{33}$, $\nu_{12} \approx \nu_{13}$ and $G_{12} \approx G_{13}$. Fig. 11 shows the relative orthotropic error norm of the stiffness matrix which has significantly reduced for random RVEs with more than eight fibres indicating the ability of the RVEs to adequately capture the general orthotropic nature of the FRP composites.

4.2. Effect on the critical strain invariants

The influence of the RVE sizes and shapes in a dehomogenisation based multiscale analysis is analysed in this section by estimating the critical strain invariants according to the process outlined in Section 2.2, which is further detailed as follows. First, the global strain tensor ($\bar{\epsilon}$) data obtained from the continuum modelling of off-axis tensile tests such as those listed in Table A2 are amplified by matrix multiplication of the MMEs probed at interrogation points to obtain the local strain tensors (ϵ) observable in the micro-scale using Eq. (1). It should be noted that each RVE model is assumed to be oriented in the range $0^\circ \leq \theta < 180^\circ$ at 15° increments performing 12 strain tensor rotations. This is to enable the global strain tensors to be applied at different orientations in the transverse plane to account for the range of possible orientations of the observed RVEs. The local strain tensors thus obtained at each of the interrogation points and orientations in an RVE are then used to compute the critical dilatational and distortional strain invariants according to Eqs. (2) and (3). As essentially a reliability study to predict the damage onset in each RVE, only the maximum value of the dilatational and the distortional strain invariant for a particular instance of the random RVE is of interest from all the interrogation points in the RVE, across all different orientations. This process is repeated for all the random RVE models with different shapes and sizes and unit cells.

Fig. 12 illustrates the trend in the observed maximum of the dilatational and distortional strain invariants for a range of RVE models and aspect ratios. The average of ten trials for each RVE size is provided with error bands indicating the 95% confidence interval of the average. The

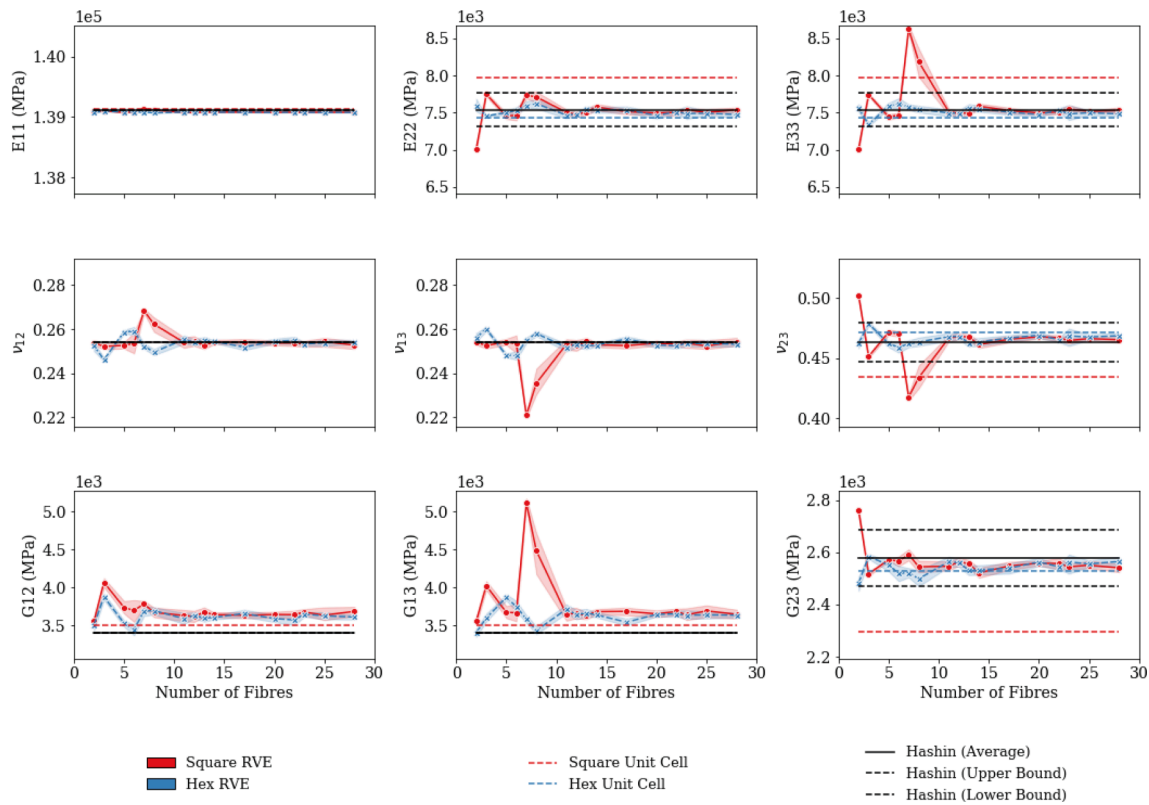


Fig. 10. Variation of homogenised elastic properties with RVE sizes.

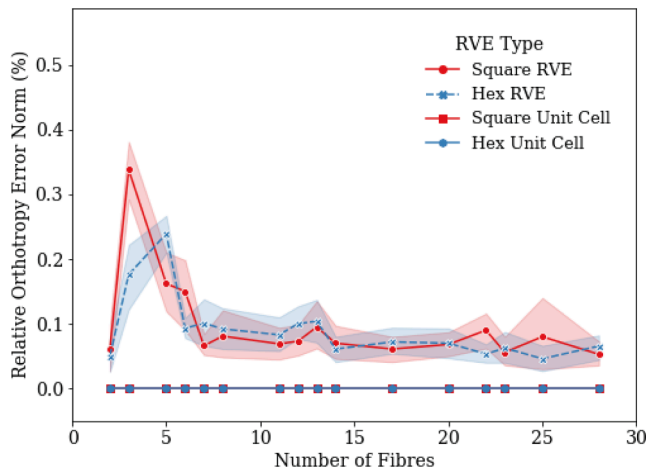


Fig. 11. Variation of relative orthotropic error norm of the stiffness matrix with RVE sizes.

critical strain invariants estimated using unit cells are also plotted as data points for comparison. It can readily be seen that the critical invariants estimated using the random RVE models containing more than two fibres are higher than both the square and hexagonal configurations of unit cell models. This clearly indicates that the random RVE models amplify the continuum strain tensor differently to the commonly used square and hexagonal unit cells, which are insufficient to accurately capture the likely variability in local field values such as strain invariants. The strain invariants exhibit considerable differences in mean values for smaller RVE sizes although they have converged within 5% beyond 11 fibre models indicating that having large number of fibres in the RVE model does not necessarily lead to better reliable observations. Moreover, the RVE aspect ratios have had negligible influence on the

dilatational strain invariants, but the distortional strain invariants exhibit considerable differences in mean values between the different aspect ratios of the same RVE size.

4.3. Effects on the damage prediction

Fig. 13 shows the damage onset surface in strain-invariant space with extracted critical invariants for all the 320 random RVE models (2 aspect ratios $\times$ 16 RVEs with different number of fibres $\times$ 10 trials) grouped into various off-axis test angles and aspect ratios. Superimposed as markers of the same colour with black borders on the figure are the critical invariants estimated from the unit cell models. The regions enclosed by the dotted lines and dashed lines correspond to the damage onset envelope predicted from the random RVE models and unit cell models, respectively. Irrespective of the RVE model utilised in the dehomogenisation of the continuum tensor to the matrix constituent, the distinct non-interacting nature of the damage initiation modes in the resin phase postulated by Onset Theory is evident. However, it should be noted that the results in Fig. 13 do not necessarily indicate that using a random RVE model is inherently more conservative or anti-conservative than the unit cells. The general dehomogenisation framework (see Fig. 4 for the one utilised in Onset Theory) uses the same RVEs or unit cells for calculating dehomogenised local critical parameters and also while predicting damage onset, i.e. both the numerator and denominator in Eq. (4) are dehomogenised using the same set of RVEs or unit cells. As such, systematic differences between individual unit cells or random RVEs are likely to produce only small effects on damage onset prediction for most load cases. Rather, variation within a set of random RVE models will produce a spread of predictions which are important to consider.

The comparison between the classes of random RVEs (sizes and shapes) and unit cell models when extended to damage onset prediction using Onset Theory is studied further by performing continuum simulation of two different laminate configurations of open-hole coupon under tensile loading: (1) $[90]_8$, (2) $[+45/-45]_4$. Elastic properties of

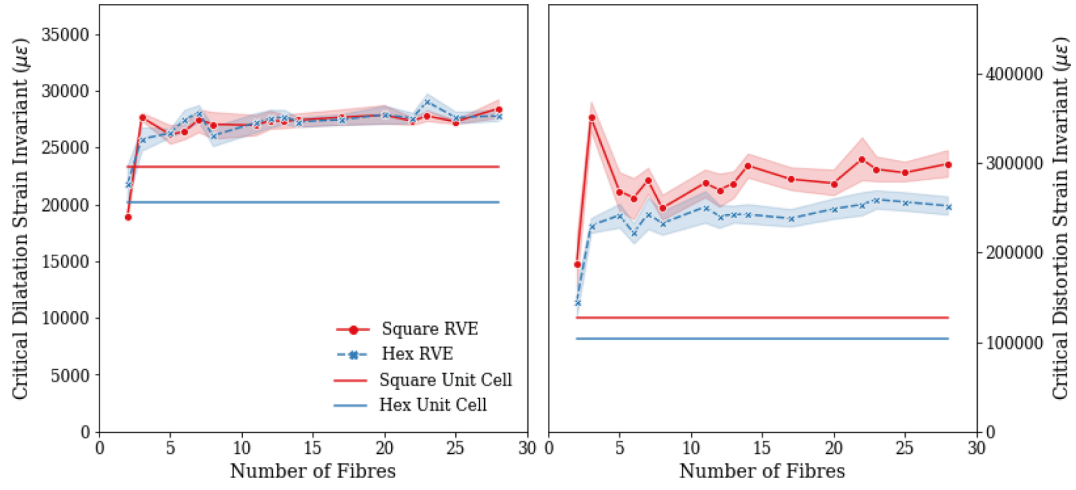


Fig. 12. Variation of strain invariants for RVEs with different numbers of fibres.

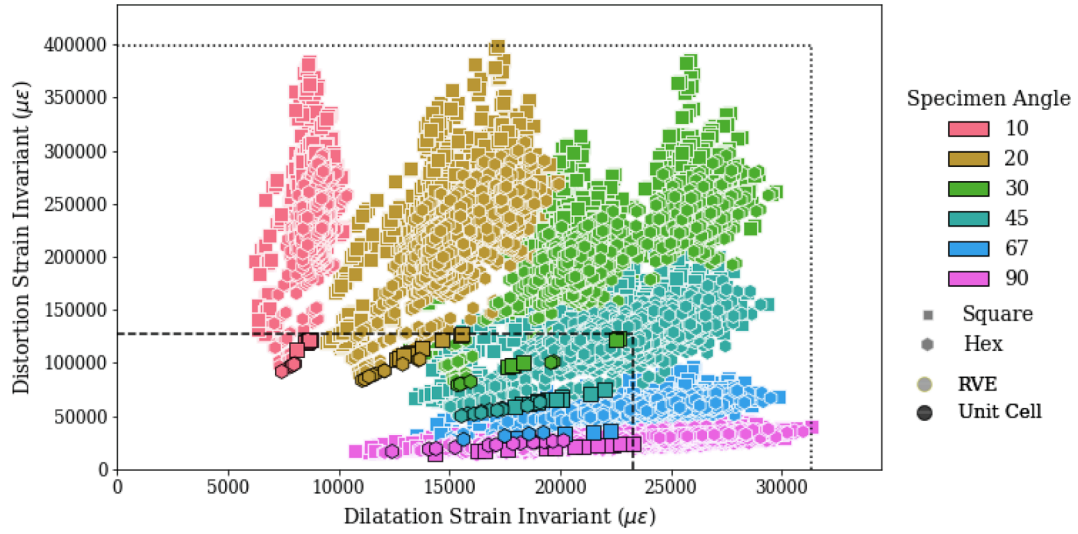


Fig. 13. Plot of $\epsilon_{dil,cr}$ vs $\epsilon_{dis,cr}$ for various off-axis tension tests predicted by RVEs.

T300/CYCOM970 prepreg which are used in the continuum simulation of open-hole tension tests are given in Table A1. The MME data obtained for the converged RVE models with sizes ranging from 11 fibres to 28

fibres and unit cells are utilised in the Abaqus UMAT subroutine incorporating the Onset Theory framework (demonstrated in [63]) to conduct the multiscale analysis of the continuum model. The strain to damage

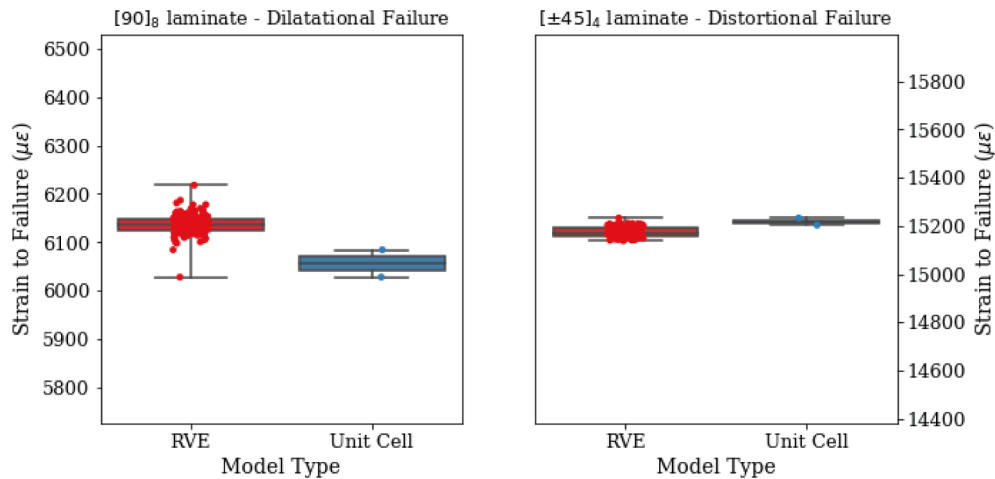


Fig. 14. Damage strains estimated by RVEs vs unit cells for OHT specimens: (left) $[90]_8$, (right) $[\pm 45]_4$.

initiation predicted by the random RVEs and unit cells in the laminate configurations of $[90]_8$ and $[+45/-45]_4$ under dilatational and distortional modes respectively is shown as box plot in Fig. 14. As expected, it is seen that the strain estimated by RVE models in the distribution in both the laminates is similar to that of the unit cell models with less than 3% difference in strain to damage initiation.

5. Discussion

It is an important research question to find the appropriate RVE size in terms of the number of fibres to be used for estimating homogenised elastic properties or dehomogenised critical strength properties using any micromechanical analysis methodology as the randomness of the fibre arrangements in the RVE can lead to variability in predictions. However, such RVE models should only capture the true variability of these properties and not be affected by artificial model-based considerations such as model domain size or the shape. On the contrary, the computational expense of such multiscale modelling approaches increases as the size of the 3D RVE increases and hence establishing the appropriate bound for the number of fibres can improve the computational efficiency of the technique.

In this work, this is achieved by conducting Monte Carlo simulations of reasonably sized RVE models and observing the geometric descriptors and statistical functions of the RVE models viz. gap ratio, Ripley's K function and 2-point correlation function and the modelling predictions in terms of the effective stiffness properties, local strain invariants and strain to damage initiation. According to the criteria recommended by Trias [51], each of these measures has helped identify a minimum RVE size in terms of the number of fibres and, consequently, the largest of these values should be used as a valid statistical volume element model.

From the range of RVE models considered in this work, the statistical analysis of microstructures and the estimated effective properties and local strain invariants have indicated that RVEs differ considerably from unit cells, and they should consist of at least 12 fibres to reproduce the similar converged statistics. For a square RVE with 60% fibre volume fraction, this translates to a δ ratio ($\delta = L/r$) of 8 which is in line with the findings of other limited works in the literature that have studied the homogenised elastic properties using 3D RVEs [23,38]. However, the aspect ratio of RVEs leads to considerable differences when local behaviour is of interest as in the case of damage analysis. Particularly, square RVEs experience a significantly higher value of distortional strain invariant than the hex RVEs for the small fibre numbers considered in this work potentially due to the differences in the fibre packing arrangements affecting shear strain terms. This difference should likely disappear for larger RVE sizes such as the 200 fibre models proposed in the literature for 2D RVEs, however at the cost of significantly increased computational demands. Yet, this has not translated into differences in damage prediction as the RVEs are used in calculating both dehomogenised/local critical parameters and design/working parameters. In future, if critical material properties such as invariants are instead measured directly from neat resin samples, the difference between the different RVE approaches is pivotal for the following reasons: (1) the unit cells will always produce anti-conservative predictions relative to random RVEs, and (2) when smaller RVEs are desired as SVEs, one should also consider the aspect ratio of the RVE domain as it could influence the properties of interest.

Nevertheless, the generalisability of these results is subject to certain limitations owing to the scope of the study. First, it should be noted that the RVEs modelled here are assumed to have circular fibres with constant fibre diameter, no matrix voids or pre-cracks and perfect fibre-matrix bonding interface. If these microstructural features are adopted in the RVE to be otherwise, they would further influence the critical size of the RVE but will still likely produce conservative results than unit cells. Secondly, the influence of fibre volume fraction of the RVE is not explored in this study. The reduced packing constraints of lower fibre volume fractions would likely increase the scatter in

statistical features and property predictions for smaller RVEs but its implication on the critical RVE size and shape should be carefully considered. Thirdly, the fibre failure modes due to clustering of fibres and the change in fibre centroid locus along the fibre direction are likely to have an additional effect on RVE size and were not investigated in this work. Where a highly robust multiscale study and its commercial design implementation considering these three factors are required, the computational cost associated is significant. This can be alleviated through reduced order models or surrogate models which are currently being pursued by the authors. Lastly, the generated RVE statistics could be compared with real microstructure statistics of the material system and experimental data can be used to compare the RVE model predictions. A further study is therefore planned that would act as a validation of the micro-mechanical analysis results and aid in demonstrating the robustness of the approach for design certifications.

6. Conclusions

A multiscale modelling technique involving random 3D RVE models of UD FRP composites was presented in this paper and comparisons with unit cell models were made primarily in the context of predicting homogenised effective properties and localised damage onset. The paper described an algorithmic way of generating pseudo-random RVE models with an arbitrary number of fibres at the micro-scale and with varying aspect ratios. Extension to a dehomogenisation based multi-scale modelling framework such as Onset Theory to utilise random 3D RVE models was also presented making use of the 2D Voronoi Tessellation approach. Statistical analyses on random RVE models using geometric descriptors and statistical functions such as nearest neighbour distance, Ripley's K function and 2-point correlation function informed the likely fibre arrangements that could lead to representative scenarios of the real lamina. Monte Carlo simulations of random RVE models with a range of fibre numbers and two aspect ratios were conducted to estimate homogenised effective properties and dehomogenised strain invariants and compared with the results of unit cell models. For the microstructural features adopted in this work, it was shown that SVE models with more than 11 fibres should be considered to efficiently capture the properties of interest in both homogenisation and dehomogenisation based multi-scale modelling frameworks. The scenarios where unit cell models would agree with or be anti-conservative in their damage initiation predictions with such appropriately sized SVEs were highlighted. A robust framework for appropriately identifying and using random RVEs is thus constructed, which can be used in a high-fidelity multi-scale reliability study of the UD FRP composites with reasonably regulated computational power.

CRedit authorship contribution statement

N.K. Balasubramani: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **B. Zhang:** Methodology, Investigation, Writing – original draft. **N.T. Chowdhury:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Visualization, Supervision. **A. Mukkavilli:** Resources, Data curation. **M. Suter:** Writing – review & editing, Supervision, Project administration. **G.M. Pearce:** Conceptualization, Methodology, Software, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the Australian Research Council (ARC) through ARC Linkage Project LP150100653. The authors would like to acknowledge the strong financial support of the ARC and the research partners at Boeing Aerostructures Australia, Boeing Defence Australia and The Boeing Company.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

Appendix A. Experimental and material data

See Tables A1–A2. (Not export controlled. Not subject to Australian or International export legislation.).

References

- [1] Kaddour AS, Hinton MJ. Maturity of 3D failure criteria for fibre-reinforced composites: Comparison between theories and experiments: Part B of WWFE-II. *J Compos Mater* 2013;47:925–66. <https://doi.org/10.1177/0021998313478710>.
- [2] Kwon YW, Allen DH, Talreja R. Multiscale modeling and simulation of composite materials and structures. Springer; 2008.
- [3] Talreja R, Singh CV. Multiscale modeling for damage analysis. *Multiscale Modeling and Simulation of Composite Materials and Structures*. Springer; 2008. p. 529–78.
- [4] Chowdhury NT, Chiu WK, Wang J, Yan W. Predicting matrix failure in composite structures using a hybrid failure criterion. *Compos Struct* 2016;137:148–58.
- [5] Li S. On the unit cell for micromechanical analysis of fibre-reinforced composites. In: *Proceedings of the Royal Society of London Series A: Mathematical, Physical and Engineering Sciences* 1999;455:815–38.
- [6] Huang W, Xu R, Yang J, Huang Q, Hu H. Data-driven multiscale simulation of FRP based on material twins. *Compos Struct* 2021;256:113013. <https://doi.org/10.1016/j.compstruct.2020.113013>.
- [7] Hill R. Elastic properties of reinforced solids: Some theoretical principles. *J Mech Phys Solids* 1963;11:357–72. [https://doi.org/10.1016/0022-5096\(63\)90036-X](https://doi.org/10.1016/0022-5096(63)90036-X).
- [8] Ha SK, Jin KK, Huang Y. Micro-Mechanics of Failure (MMF) for Continuous Fiber Reinforced Composites. *J Compos Mater* 2008;42:1873–95. <https://doi.org/10.1177/0021998308093911>.
- [9] Pearce G, Mukkavilli A, Chowdhury NT, Lim SH, Prusty G, Crosky A, et al. Strain Invariant Failure Theory – Part 1: An extensible framework for predicting the mechanical performance of fibre reinforced polymer composites. *Composite Struct* 2019;209:1022–34. <https://doi.org/10.1016/j.compstruct.2018.03.084>.
- [10] Asp LE, Berglund LA, Talreja R. A criterion for crack initiation in glassy polymers subjected to a composite-like stress state. *Compos Sci Technol* 1996;56:1291–301. [https://doi.org/10.1016/S0266-3538\(96\)00090-5](https://doi.org/10.1016/S0266-3538(96)00090-5).
- [11] Garnich MR, Hansen AC. A Multicontinuum Theory for Thermal-Elastic Finite Element Analysis of Composite Materials. *J Compos Mater* 1997;31:71–86. <https://doi.org/10.1177/002199839703100105>.
- [12] Helius: MCT Technical Manual Version 1.0 ed 2009.
- [13] Buchanan DL, Gosse JH, Wollschlaeger JA, Ritchey A, Byron Pipes R. Micromechanical enhancement of the macroscopic strain state for advanced composite materials. *Compos Sci Technol* 2009;69:1974–8. <https://doi.org/10.1016/j.compstruct.2009.04.022>.
- [14] Jin Kyo-Kook, Lee Young-Hwan, Kyu Sung, Ha YH. Distribution of Micro Stresses and Interfacial Tensions in Unidirectional Composites. *J Compos Mater* 2008;42:1825–49.
- [15] Huang Y, Jin KK, Ha SK. Effects of fiber arrangement on mechanical behavior of unidirectional composites. *J Compos Mater* 2008;42:1851–71.
- [16] Hart-Smith LJ. Application of the strain invariant failure theory (SIFT) to metals and fiber–polymer composites. *Phil Mag* 2010;90:4263–331. <https://doi.org/10.1080/14786435.2010.510450>.
- [17] Chowdhury NT, Wang J, Chiu WK, Yan W. Matrix failure in composite laminates under tensile loading. *Compos Struct* 2016;135:61–73. <https://doi.org/10.1016/j.compstruct.2015.09.022>.
- [18] Trias D, Costa J, Mayugo JA, Hurtado JE. Random models versus periodic models for fibre reinforced composites. *Comput Mater Sci* 2006;38:316–24. <https://doi.org/10.1016/j.commatsci.2006.03.005>.
- [19] Omairey SL, Dunning PD, Sriramula S. Influence of micro-scale uncertainties on the reliability of fibre-matrix composites. *Compos Struct* 2018;203:204–16. <https://doi.org/10.1016/j.compstruct.2018.07.026>.
- [20] Melro AR, Camanho PP, Pinho ST. Influence of geometrical parameters on the elastic response of unidirectional composite materials. *Composite Struct* 2012;94:3223–31.
- [21] Vaughan TJ, McCarthy CT. Micromechanical modelling of the transverse damage behaviour in fibre reinforced composites. *Compos Sci Technol* 2011;71:388–96.
- [22] Pulungan D, Lubineau G, Yudhanto A, Yaldiz R, Schijve W. Identifying design parameters controlling damage behaviors of continuous fiber-reinforced thermoplastic composites using micromechanics as a virtual testing tool. *Int J Solids Struct* 2017;117:177–90.
- [23] Wu L, Chung CN, Major Z, Adam L, Noels L. From SEM images to elastic responses: A stochastic multiscale analysis of UD fiber reinforced composites. *Compos Struct* 2018;189:206–27. <https://doi.org/10.1016/j.compstruct.2018.01.051>.
- [24] Koley S, Mohite PM, Upadhyay CS. A micromechanical study and uncertainty quantification for effective properties of unidirectional fibre reinforced composites. *Compos Struct* 2019;225:111141.
- [25] Hojo M, Mizuno M, Hobbiebrunken T, Adachi T, Tanaka M, Ha SK. Effect of fiber array irregularities on microscopic interfacial normal stress states of transversely loaded UD-CFRP from viewpoint of failure initiation. *Compos Sci Technol* 2009;69:1726–34. <https://doi.org/10.1016/j.compstruct.2008.08.032>.
- [26] Bostanabad R, Zhang Y, Li X, Kearney T, Brinson LC, Apley DW, et al. Computational microstructure characterization and reconstruction: Review of the state-of-the-art techniques. *Prog Mater Sci* 2018;95:1–41.
- [27] Melro A, Camanho P, Pinho S. Generation of random distribution of fibres in long-fibre reinforced composites. *Compos Sci Technol* 2008;68:2092–102. <https://doi.org/10.1016/j.compstruct.2008.03.013>.
- [28] Ripley BD. The second-order analysis of stationary point processes. *J Appl Probability* 1976;13:255–66.
- [29] Pyrz R. Quantitative description of the microstructure of composites. Part I: Morphology of unidirectional composite systems. *Compos Sci Technol* 1994;50:197–208.
- [30] Pyrz R. Correlation of microstructure variability and local stress field in two-phase materials. *Mater Sci Eng, A* 1994;177:253–9.
- [31] Lakhal L, Brunet Y, Kanit T. Evaluation of second-order correlations adjusted with simulated annealing on physical properties of unidirectional nonoverlapping fiber-reinforced materials (UD Composites). *Int J Mod Phys C* 2019;30:1950017. <https://doi.org/10.1142/S0129183119500177>.
- [32] Lakhal L, Brunet Y, Kanit T. Numerical bounds for elastic properties of unidirectional non-overlapping fiber reinforced materials. *Eur J Mech A Solids* 2020;80:103926. <https://doi.org/10.1016/j.euromechsol.2019.103926>.
- [33] Torquato S. *Random Heterogeneous Materials: Microstructure and Macroscopic Properties*. New York: Springer; 2002.
- [34] Kanit T, Forest S, Galliet I, Mounoury V, Jeulin D. Determination of the size of the representative volume element for random composites: statistical and numerical approach. *Int J Solids Struct* 2003;40:3647–79. [https://doi.org/10.1016/S0020-7683\(03\)00143-4](https://doi.org/10.1016/S0020-7683(03)00143-4).
- [35] Wongsto A, Li S. Micromechanical FE analysis of UD fibre-reinforced composites with fibres distributed at random over the transverse cross-section. *Compos A Appl Sci Manuf* 2005;36:1246–66. <https://doi.org/10.1016/j.compositesa.2005.01.010>.
- [36] López Jiménez F. On the isotropy of randomly generated representative volume elements for fiber-reinforced elastomers. *Compos B Eng* 2016;87:33–9. <https://doi.org/10.1016/j.compositesb.2015.10.014>.
- [37] Pathan MV, Tagarielli VL, Patsias S. Effect of fibre shape and interphase on the anisotropic viscoelastic response of fibre composites. *Composite Struct* 2017;162:156–63.
- [38] Okereke MI, Akpoyomare AI. A virtual framework for prediction of full-field elastic response of unidirectional composites. *Computational Mater Sci* 2013;70:82–99.
- [39] Saoudi T, El-Moumen A, Kanit T, Belouchrani MEA, Benseddig N, Imad A. Numerical Evaluation of the Thermal Properties of UD-Fibers Reinforced Composites for Different Morphologies. *Int J Appl Mech* 2020;12:2050032. <https://doi.org/10.1142/S1758825120500325>.
- [40] Bulsara VN, Talreja R, Qu J. Damage initiation under transverse loading of unidirectional composites with arbitrarily distributed fibers. *Compos Sci Technol* 1999;59:673–82.
- [41] Elnekhaily SA, Talreja R. Damage initiation in unidirectional fiber composites with different degrees of nonuniform fiber distribution. *Compos Sci Technol* 2018;155:22–32.
- [42] González C, Llorca J. Mechanical behavior of unidirectional fiber-reinforced polymers under transverse compression: Microscopic mechanisms and modeling. *Compos Sci Technol* 2007;67:2795–806.
- [43] Vajari DA, González C, Llorca J, Legartha BN. A numerical study of the influence of microvoids in the transverse mechanical response of unidirectional composites. *Compos Sci Technol* 2014;97:46–54.
- [44] Yang L, Yan Y, Ma J, Liu B. Effects of inter-fiber spacing and thermal residual stress on transverse failure of fiber-reinforced polymer–matrix composites. *Comput Mater Sci* 2013;68:255–62.
- [45] Yang Lei, Ma Jian, Bo Liu YY. Effects of inter-fiber spacing and thermal residual stress on transverse failure of fiber-reinforced polymer–matrix composites. *Comput Mater Sci* 2013;68:255–62.
- [46] Okabe T, Imamura H, Sato Y, Higuchi R, Koyanagi J, Talreja R. Experimental and numerical studies of initial cracking in CFRP cross-ply laminates. *Compos A* 2015;68:81–9.
- [47] Ghayoor H, Hoa SV, Marsden CC. A micromechanical study of stress concentrations in composites. *Compos Part B: Eng* 2018. <https://doi.org/10.1016/j.compositesb.2017.09.009>.
- [48] Sun Q, Zhou G, Meng Z, Guo H, Chen Z, Liu H, et al. Failure criteria of unidirectional carbon fiber reinforced polymer composites informed by a computational micromechanics model. *Compos Sci Technol* 2019;172:81–95.
- [49] López Jiménez F. Variations in the distribution of local strain energy within different realizations of a representative volume element. *Compos B Eng* 2019;176:107111. <https://doi.org/10.1016/j.compositesb.2019.107111>.
- [50] Oh JH, Jin KK, Ha SK. Interfacial strain distribution of a unidirectional composite with randomly distributed fibers under transverse loading. *J Compos Mater* 2006;40:759–78.

- [51] Trias D, Costa J, Turon A, Hurtado J. Determination of the critical size of a statistical representative volume element (SRVE) for carbon reinforced polymers. *Acta Materialia* 2006;54:3471–84.
- [52] Romanov V, Lomov SV, Swolfs Y, Orlova S, Gorbatiikh L, Verpoest I. Statistical analysis of real and simulated fibre arrangements in unidirectional composites. *Compos Sci Technol* 2013;87:126–34.
- [53] Sudhir A, Talreja R. Simulation of manufacturing induced fiber clustering and matrix voids and their effect on transverse crack formation in unidirectional composites. *Compos A Appl Sci Manuf* 2019;127:105620.
- [54] Niezgoda SR, Turner DM, Fullwood DT, Kalidindi SR. Optimized structure based representative volume element sets reflecting the ensemble-averaged 2-point statistics. *Acta Materialia* 2010;58:4432–45.
- [55] Tian W, Qi L, Zhou J, Liang J, Ma Y. Representative volume element for composites reinforced by spatially randomly distributed discontinuous fibers and its applications. *Compos Struct* 2015;131:366–73. <https://doi.org/10.1016/j.compstruct.2015.05.014>.
- [56] Park SM, Lim JH, Seong MR, Sohn D. Efficient generator of random fiber distribution with diverse volume fractions by random fiber removal. *Compos B Eng* 2019. <https://doi.org/10.1016/j.compositesb.2018.12.042>.
- [57] Yang L, Yan Y, Ran Z, Liu Y. A new method for generating random fibre distributions for fibre reinforced composites. *Compos Sci Technol* 2013;76:14–20.
- [58] Pathan MV, Tagarielli VL, Patsias S, Baiz-Villafranca PM. A new algorithm to generate representative volume elements of composites with cylindrical or spherical fillers. *Compos Part B: Eng* 2017. <https://doi.org/10.1016/j.compositesb.2016.10.078>.
- [59] Beicha D, Kanit T, Brunet Y, Imad A, el Moumen A, Khelfaoui Y. Effective transverse elastic properties of unidirectional fiber reinforced composites. *Mech Mater* 2016;102:47–53. <https://doi.org/10.1016/j.mechmat.2016.08.010>.
- [60] Gosse JH, Christensen S. Strain invariant failure criteria for polymers in composite materials. In: 19th AIAA Applied Aerodynamics Conference. Anaheim, CA, USA: American Institute of Aeronautics and Astronautics; 2001. p. 45–55. <https://doi.org/10.2514/6.2001-1184>.
- [61] Garoz D, Gilabert FA, Sevenois RDB, Spronk SWF, van Paepegem W. Consistent application of periodic boundary conditions in implicit and explicit finite element simulations of damage in composites. *Compos B Eng* 2018;168:254–66.
- [62] Gosse RBP, Gosse JH. An onset theory for irreversible deformation in composite materials. In: 17th International Conference on Composite Materials 2009.
- [63] Chowdhury NT, Balasubramani NK, Pearce GM, Tao C. A multiscale modelling procedure for predicting failure in composite textiles using an enhancement approach. *Eng Fail Anal* 2019;102:148–59. <https://doi.org/10.1016/j.engfailanal.2019.04.013>.
- [64] Pitz EJ, Miron M-C, Major Z. Computational implementation and experimental validation of a micro-mechanics informed progressive damage Strain Invariant Failure Theory. *Compos B Eng* 2018;155:225–36. <https://doi.org/10.1016/j.compositesb.2018.08.040>.
- [65] Mukkavilli A. The Effects of Temperature on the Onset of Failure in Polymer Matrix Composites. University of New South Wales; 2019.
- [66] Hashin Z. Analysis of Properties of Fiber Composites With Anisotropic Constituents. *J Appl Mech* 1979;46:543–50. <https://doi.org/10.1115/1.3424603>.